DATASCI 406 FA 26, Quiz 1.5

Name and Uniqname:

Question 1 (4 points)

Using today's exercises, identify an example of each of these techniques we have covered in this module.

(a) Estimating a feature of a distribution (such as an expectation or probability) for which you do not have a closed form answer.

(b) Finding the operating characteristic of an estimator or testing procedure.

(c) Approximating the sampling distribution of a statistic that has no tractable form.

(d) Investigating a claim you have no formal proof of.

Question 2 (1 point)

The paper's conclusion mentions tests built on the **second** digit rather than the first. To run today's test that way you would need the distribution of the second digit under the null.

Assume Benford's Law still holds — that is, $F \sim U(0, 1)$. Explain how you would use a Monte Carlo technique to estimate the probability of second digits $d_2 = 0, \ldots, 9$. You may assume a function `second_digit()` exists already.